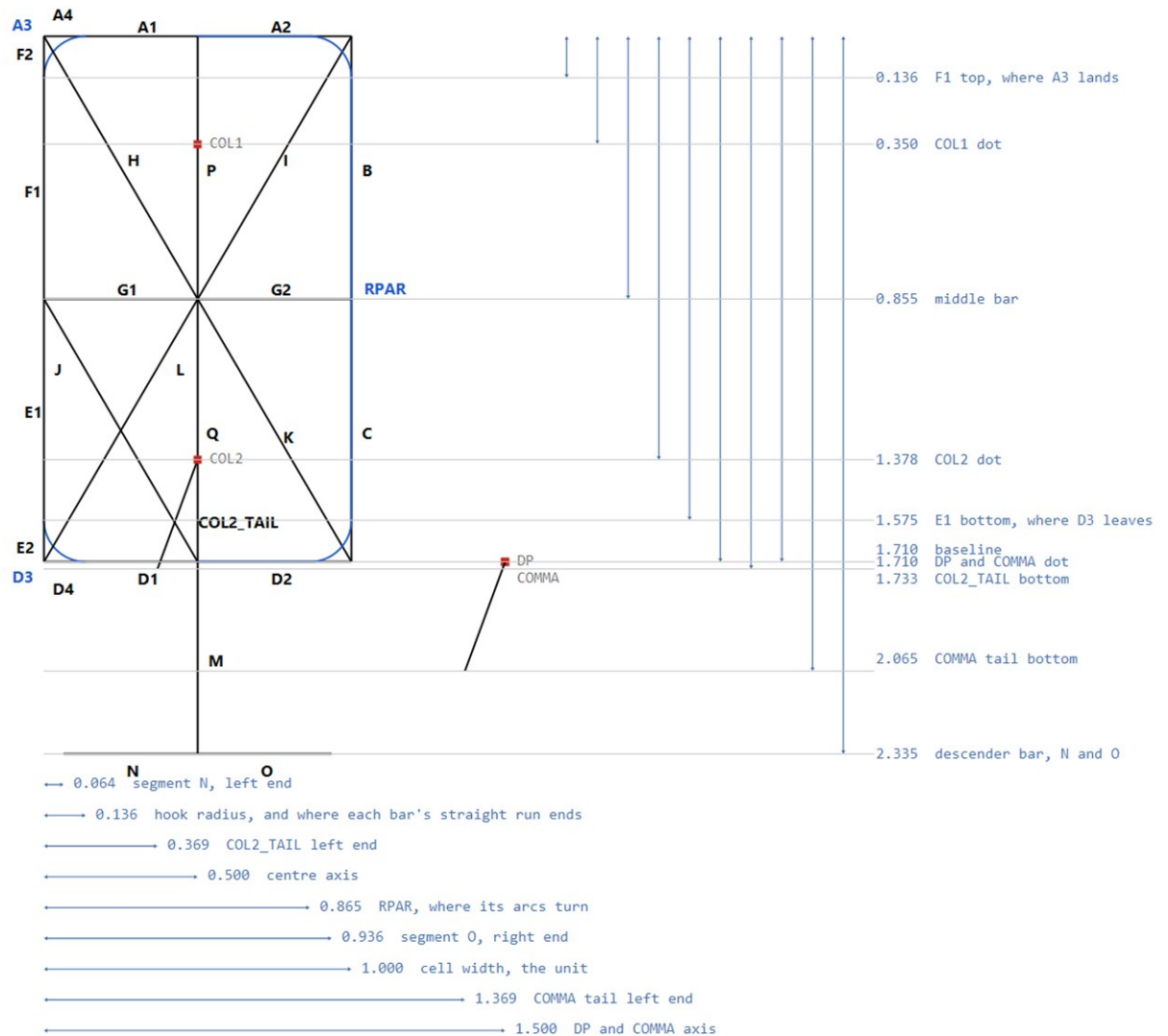


TalkRPN cell - centreline skeleton

A 1970s bubble-LED look: HP-01 styling (rounded left corners) on a modified DL-3422 segment set, so the display can show text as well as digits. Identical to neither part. Segments meet flush where real hardware leaves gaps, and the decimal point sits in the gap after the cell rather than taking a cell of its own.

THE UNIT: segment E/F to segment B/C is 1 - the left column to the right column, centre to centre. Every length here is in that unit, across and down alike, to four significant figures, so the centre axis is exactly 0.5 and the gap reads directly as the clearance between neighbours.

No stroke width, no slant: these are centrelines, and both are applied at render time. Origin is the top-left centreline corner; x right, y down. 32 elements, so the mask is a Long. Blue: A3/D3 are ALTERNATIVES to A4/D4, never both lit; RPAR is the whole right paren as one element. TalkRpnFont.kt is the source of truth - this sheet mirrors it.



TalkRPN cell - centreline listing

SEGMENT	CENTRELINES	from	to	
A1	line	0.500, 0.000	-> 0.136, 0.000	
A2	line	0.500, 0.000	-> 1.000, 0.000	
A4	line	0.136, 0.000	-> 0.000, 0.000	
A3	arc	0.136, 0.000	-> 0.000, 0.136	arc R0.136 centre 0.136,0.136
B	line	1.000, 0.000	-> 1.000, 0.855	
C	line	1.000, 0.855	-> 1.000, 1.710	
D1	line	0.500, 1.710	-> 0.136, 1.710	
D2	line	0.500, 1.710	-> 1.000, 1.710	
D4	line	0.136, 1.710	-> 0.000, 1.710	
D3	arc	0.136, 1.710	-> 0.000, 1.575	arc R0.136 centre 0.136,1.575
F2	line	0.000, 0.000	-> 0.000, 0.136	
F1	line	0.000, 0.136	-> 0.000, 0.855	
E1	line	0.000, 0.855	-> 0.000, 1.575	
E2	line	0.000, 1.575	-> 0.000, 1.710	
G1	line	0.000, 0.855	-> 0.500, 0.855	
G2	line	0.500, 0.855	-> 1.000, 0.855	
H	line	0.000, 0.000	-> 0.500, 0.855	
I	line	1.000, 0.000	-> 0.500, 0.855	
J	line	0.000, 0.855	-> 0.500, 1.710	
L	line	0.500, 0.855	-> 0.000, 1.710	
K	line	0.500, 0.855	-> 1.000, 1.710	
P	line	0.500, 0.000	-> 0.500, 0.855	
Q	line	0.500, 0.855	-> 0.500, 1.710	
COL2_TAIL	line	0.500, 1.378	-> 0.369, 1.733	
M	line	0.500, 1.710	-> 0.500, 2.335	
N	line	0.064, 2.335	-> 0.500, 2.335	
O	line	0.500, 2.335	-> 0.936, 2.335	
RPAR	compound	0.500,0.000	-> arc -> 1.000, 0.136..1.575 -> arc -> 0.500,1.710	the whole right paren

DOT CENTRES

COL1	0.500, 0.350	upper colon dot; also dots i and j
COL2	0.500, 1.378	lower colon dot, colon only
DP	1.500, 1.710	decimal point, outside the cell to the right
COMMA	1.500, 1.710	same centre, plus a tail down and left
SQUARE	side = 2 x STROKE = 0.295	

CENTRELINE BOX 1.000 wide x 1.710 tall aspect 0.585
2.335 tall including the descender

GAP 1.000 last centreline of one glyph to the first of the next; the ink meets at one stroke
VPITCH floor is 2.483, the ink height - below that descenders reach the next row
SLANT 6.0 degrees, applied at render

RELATIONSHIPS

hook radius = hook start x = 0.136, so each arc is tangent to both bar and column
F1 top = 0.136 = where A3 lands. E1 bottom = 1.575 = where D3 lands.
F2 and E2 are the stubs that carry the left column out to the corner when it is square.
B meets C at y = 0.855: the right column is undivided at the corners, which is what makes a 4 lopsided.
N and O are inset 0.0640 and 0.0641 from the columns - symmetric to within measurement.
COL2_TAIL and the COMMA tail are the SAME tail, 0.355 down and 0.131 left, hung off two different dots.
That is what makes a semicolon a colon whose lower dot grew a tail, at no cost in new geometry.

THE THREE LEFT-COLUMN ENDINGS

square	A4 + F2	most letters
hooked	A3 + F1	0 2 3 5 7 8 9, A C G O Q S, (& and @ (F2 dark)
short	F1 alone	digit 4, no A at all - left side sits 0.136 lower than the right

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
2																	
	sp	! P COL2	" F2 F1 P	# A4 A1 A2 G1 G2 N O P Q H	\$ A3 A1 A2 F1 G1 G2 C D4 D1 D2 P Q H	% A3 A1 F1 G1 P I L G2 C D2 Q	& A4 A1 P H G1 E1 D3 D1 D2 C K	' I	(A1 A3 F1 E1 D3 D1	) RPAR	* H I L K P Q G1 G2	+ G1 G2 P Q	, COMMA	- G1 G2	. DP	/ I L	
3																	
	0 A3 A1 A2 F1 B E1 C G3 D1 D2	1 B C	2 A3 A1 A2 B G1 G2 E1 D3 D1 D2	3 A3 A1 A2 B G1 G2 C D3 D1 D2	4 F1 G1 G2 B C	5 A3 A1 A2 F1 G1 G2 C D3 D1 D2	6 A4 A1 A2 F2 F1 G1 G2 E1 E2 C D4 D1 D2	7 A3 A1 A2 B C	8 A3 A1 A2 F1 B G1 G2 E1 C D3 D1 D2	9 A3 A1 A2 F1 B G1 G2 C D3 D1 D2	:	COL1 COL2	; COL1 COL2 COL3_TAIL	< I K	= G1 D4 D1	> H L	? A4 A1 A2 B G2 Q
4																	
	@ A3 A1 A2 B C D3 D1 D2 G1 G1 Q	A A3 A1 A2 F1 B G1 G2 E1 E2 C	B A4 A1 A2 B C D4 D1 D2 G2 P Q	C A3 A1 A2 F1 E3 D3 D1 D2	D A4 A1 A2 B C D4 D1 D2 P Q	E A4 A1 A2 F2 F1 G1 E1 E2 D4 D1 D2	F A4 A1 A2 F2 F1 G1 E1 E2	G A3 A1 A2 F1 E1 D3 D1 D2 C G2	H F2 F1 B G1 G2 E1 E2 C	I A4 A1 A2 P Q D4 D1 D2	J B C E1 D3 D1 D2	K F2 F1 E1 E2 G1 I K	L F2 F1 E1 E2 D4 D1 D2	M F2 F1 H I B E1 E2 C	N F2 F1 H B K E1 E2 C	O A4 A1 A2 F2 F1 B E1 E2 C D4 D1 D2	
5																	
	P A4 A1 A2 F2 F1 B G1 G2 E1 E2	Q A3 A1 A2 F1 B E1 C D3 D1 D2 K	R A4 A1 A2 F2 F1 B G1 G2 E1 E2 K	S A3 A1 A2 F1 G1 G2 C D4 D1 D2	T A4 A1 A2 P Q	U F2 F1 B E1 C D3 D1 D2	V F2 F1 E1 E2 I L	W F2 F1 B E1 E2 C L K	X H I L K	Y H I Q	Z A4 A1 A2 I L D4 D1 D2	[A2 P Q H O	\ H K	] A1 P Q H N	^ L K	_ D4 D1 D2	
6																	
	~ H	a G1 Q D4 D1 L	b F2 F1 E1 G1 D3 D1 Q	c E1 G1 D3 D1	d E1 G1 D3 D1 P Q	e E1 E2 G1 D4 D1 L	f A2 P Q G1 G2	g E1 G1 D3 D1 Q H N	h F2 F1 E1 E2 G1 Q	i Q COL1	j Q H N COL1	k F2 F1 E1 E2 G1 J	l P Q	m G1 G2 E1 E2 C Q	n E1 E2 P Q	o E1 G1 D3 D1 Q	
7																	
	p C G2 D2 Q H	q E1 G1 D3 D1 Q H O	r E1 E2 G1	s G1 J D4 D1	t P Q G1 G2 D2	u E1 D3 D1 Q	v E1 E2 L	w E1 E2 C L K	x J L	y E1 E2 D4 D1 Q H N	z G1 L D4 D1	{ A2 P G1 H O	P Q H	} A1 P G2 Q H N	~ F1 A3 A1 P G2 B	DEL all segments	